

COMMENTARY

Trial informativeness across research practices: initial findings from a mapping exercise

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Accepted 9 April 2026; Published online 16 April 2026

Abstract

Background: Trial informativeness is influenced by multiple interacting factors across trial design, conduct, and knowledge mobilization, yet how these elements align across current research practices remains unclear. This short communication presents the initial results of a mapping exercise aimed at identifying areas of convergence across three related research strands with the goal of clarifying shared processes and actions that underpin informative trials and the broader research systems in which they operate.

Study Design and Setting: The findings from a rapid review of global insights on trial informativeness, a content analysis of internationally published clinical trial guidance documents, and qualitative interviews with trial stakeholders were compared. Actions and processes to improve trial informativeness that were verbatim or conceptually similar were extracted and organized under a shared framework. Supporting descriptors were compared and synthesized to capture core learnings that consistently appeared across all evidence sources.

Results: Comparison of the three research strands identified a set of 12 good practice actions spanning trial design, conduct, and knowledge mobilization. These actions reflect consistent priorities identified across evidence, guidance, and stakeholder perspectives. Their implementation involves multiple actors across the lifecycle of a trial. All good practice actions identified could apply to funders and/or sponsors, and trial investigators. Other stakeholders include regulatory authorities, patient or community representatives, ethics bodies, policymakers, and so on.

Conclusion: The good practice actions identified to strengthen research systems that contribute to informative trials warrant standalone consideration within a wider mapping exercise, as they address broad drivers of trial quality and support improvements both within and beyond individual studies. Future work should focus on applying, evaluating, and refining these actions to ensure they remain relevant across diverse trial settings and contexts. © 2026 The Author(s). Published by Elsevier Inc. This is an open access article under the CC BY license (<http://creativecommons.org/licenses/by/4.0/>).

Keywords: Trial informativeness; Research systems; Evidence mapping; Clinical trial design; Trial conduct; Knowledge mobilization

1. Introduction

Trial informativeness broadly refers to the extent to which a clinical trial is designed, conducted, and mobilized in ways that ensure its results are useful for decision-making and future research [1]. Despite good intentions,

many trials fail to be informative, resulting in research waste and raising ethical concerns for participants [1,2]. Shortcomings in trial design and execution are often recognized only in retrospect, with definitions of what constitutes good practice differing across stakeholder perspectives. There is a general lack of clear, actionable measures to improve informativeness in clinical trials.

To address these gaps, we undertook a mapping exercise drawing on three strands of work from the improving pre-funding peer review to increase the informativeness of randomized trials (INFORM) project: (1) a rapid review of global insights, (2) a content analysis of international trial guidance documents, and (3) interviews with diverse trial stakeholders [3–5]. This short communication summarizes the areas of convergence when the three research strands

Funding: This publication is based on research funded by the Gates Foundation, grant INV-067716. The findings and conclusions contained within are those of the authors and do not necessarily reflect positions or policies of the Gates Foundation.

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What is new?

Key findings

- Mapping across three complementary research strands identified consistent processes that underpin trial informativeness across trial design, conduct, and knowledge mobilization.
- Trial informativeness was consistently understood as context-dependent and shaped by features of the wider research system, including geography, available resources, and funding structures.

What this adds to what was known?

- These initial findings from a larger mapping exercise clarify how responsibilities and expectations can be distributed across multiple actors rather than resting solely with an individual or agency.
- The findings demonstrate how consideration of wider research systems is central to supporting informative trials, complementing existing trial-level methodological guidance.

What is the implication and what should change now?

- Improving the informativeness of trials requires greater attention to how research systems enable, support, and sustain good practices, with actions interpreted and applied in ways that are responsive to local contexts and capacities.

were compared side by side. From this process, we developed an initial set of good practice actions to support the delivery of informative trials, which provide practical considerations across trial design, conduct, and knowledge mobilization.

2. Methods

INFORM started with a rapid review of published global insights on design processes that can improve trial informativeness [4]. Global insights were defined as recommended processes or actions in the international literature intended to improve the informativeness of trials. These insights were aligned with the key conditions for trial informativeness as described by Zarin et al: importance, design, feasibility, integrity, and reporting [1,4]. From this work, a range of actions and processes that could improve trial informativeness were captured which formed the basis of qualitative analysis for both a content analysis of guiding trial documents and further interviews with trial stakeholders [3–5]. The methods underlying the three research strands

are available in their respective publications [3–5]. By aligning the coding structure across all three research outputs, adjusting as necessary for new findings, a mapping exercise was readily feasible to consolidate actions or processes to a concise, actionable list of measures to improve trial informativeness.

Mapping is a comparative method that enables the systematic alignment of findings across different sources of evidence [6]. In the first instance, actions or processes to improve trial informativeness that were verbatim or of similar concept were extracted from each research strand and organized under the respective headings from Zarin et al ([Supplementary Material 1](#)). Supporting descriptors, which varied slightly between the three strands, were then compared and synthesized by one study team member (S. P.) to capture key ideas. These insights were discussed among the study team members (S. P., S. T., M. B., and J. B.) who agreed that their utility could be best actioned if like concepts were further combined under the practical headings of trial design, conduct, and knowledge mobilization.

As implementation of good practice will vary by stakeholder, a visual roadmap was developed highlighting broad key actions or processes that funders can leverage for improved trial informativeness ([Fig](#)). This illustration is not exhaustive but rather emphasizes key areas in the life-cycle of a trial where funders can have crucial input including priority setting, design review, accountability and team support, and transparency and reporting. We present [Figure](#) from the perspective of funders because funding decisions represent a crucial point at which concerns about trial informativeness can be identified and addressed; given the space constraints of this brief communication, additional figures and materials reflecting the wider range of stakeholders will be further developed and made freely available as part of INFORM's training and dissemination package.

3. Results

[Table 1](#) presents 12 core actions for good practice to strengthen research systems for informative trials. This includes six good practice actions under the heading of design, four relevant to conduct, and two promoting knowledge mobilization. Targeted key stakeholders are also noted. All good practice actions identified could apply to funders and/or sponsors, and trial investigators. Other stakeholders include regulatory authorities, patient or community representatives, ethics bodies, policymakers, and so on. These listings recognize the diversity of individuals and agencies involved in ensuring a trial is informative and are directly reflective of the findings that have come from the mapping of the three research strands.

To aid comprehension and recall key insights, the roadmap presented in [Figure](#) suggests how some of the good

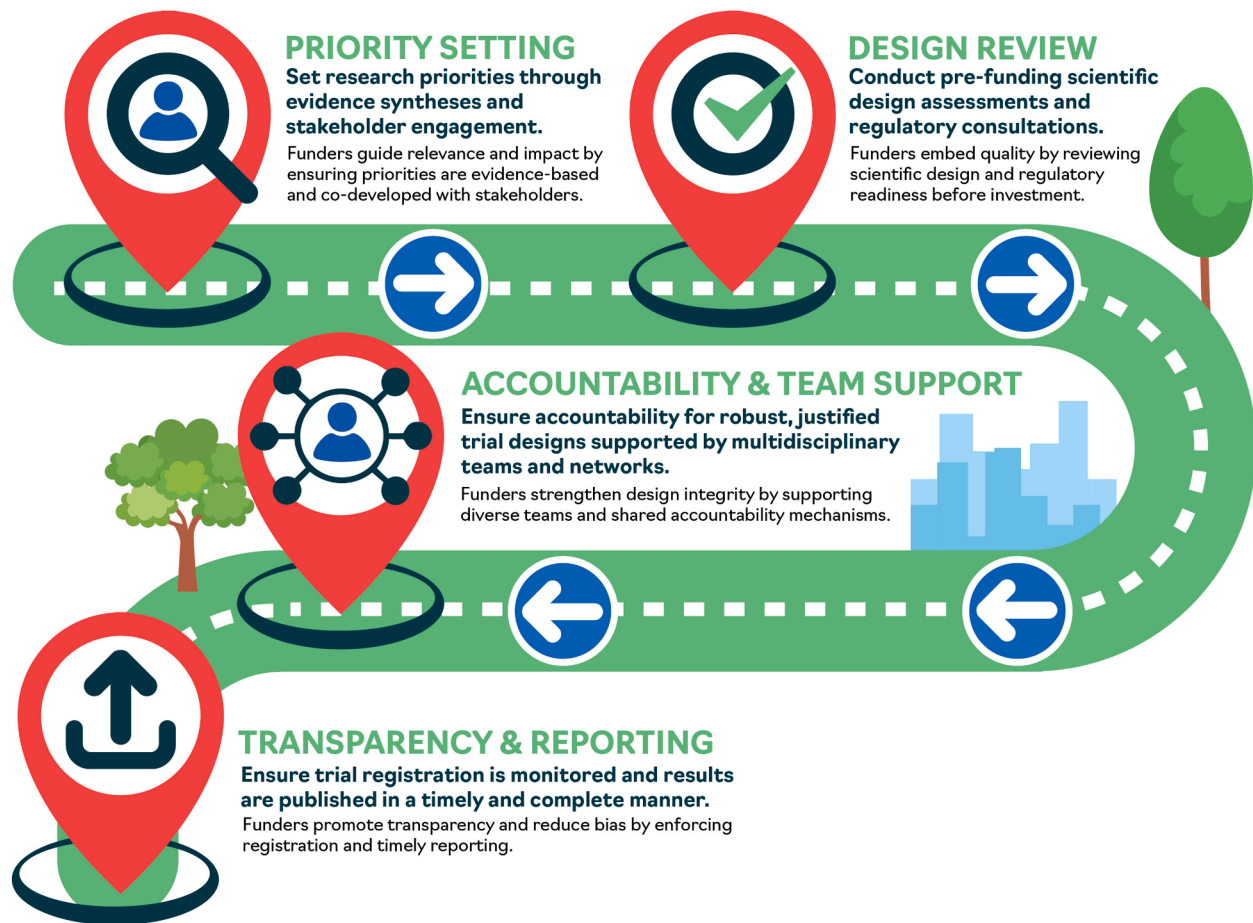


Figure. The INFORM Good Practice Actions pathway for funders: from priority setting through to transparent reporting.

practice actions could be highlighted as relevant to funding bodies. The roadmap provides a high-level overview of key stages where practices that support trial informativeness can be applied, whereas Table 1 presents the recommended actions that underpin these practices in greater detail. Informativeness was found to be context-dependent and is shaped by geography, available resources, funding structures, and other related factors. As such, not all actions to improve trial informativeness will be applicable or reasonable for all stakeholders. Informativeness is not a “one size fits all” concept, and as our three research strands have shown, approaches must be tailored to local contexts and capacities.

4. Discussion

Actions for good practice that drive informative clinical trials span design, conduct, and knowledge mobilization, reflecting the need for improved ways of working at every stage.

The good practice actions relevant to trial design must be interpreted and applied in ways that ensure they are

fit-for-purpose. For example, analysis across the INFORM research strands highlighted a consistent emphasis on collaboration with patients, local communities, other trial organizations or networks, oversight bodies, and so on. This spotlight on collaboration highlights the shared responsibility of all stakeholders in strengthening trial quality and relevance. However, as our program of research has shown, there are often disparities between the priorities and expectations of external funders and the realities of designing and implementing trials [3–5]. This is markedly relevant in discussions surrounding low-resource settings, where health care systems and infrastructure may differ substantially from those in high-income contexts [7,8]. Ensuring that trials are informative requires design processes that account for local context, including cultural and structural factors.

Trial conduct should consider all elements within the lifecycle of a trial, with a particular focus on maintaining integrity and sustainable development over time. The COVID-19 pandemic emphasized the importance of future-proofing trials by anticipating potential changes in technology and wider research practice. A recent study on future-proofing EU regulatory and market access as a result of COVID-19 describes a “need for enhanced

Table 1. Good practice actions to strengthen research systems for informative trials

Design	Key stakeholders	Rationale and implementation
1. Set research priorities through evidence syntheses and stakeholder engagement to ensure trials address recognized needs Research priorities should be informed by existing evidence and coordinated across stakeholders to ensure trials address recognized health needs and deliver meaningful benefits to target populations.	- Funder/sponsor - Investigator - Regulatory authority	Important because... Research priorities need to take account of existing evidence; this should be done in consultation with relevant stakeholders to ensure trials address recognized needs with tangible benefits for target populations. Without this... Failure to align research priorities with existing evidence and stakeholder input can result in trials focusing on areas of low relevance, leading to wasted resources and missed opportunities to address critical health needs. How this can be best done... Draw on multiple forms of existing evidence (such as systematic reviews, meta-analyses, scoping reviews, Value of information (VOI) analysis, biological knowledge, preliminary data, preclinical data); coordinating with stakeholders (such as local health care professionals; people with lived experience of a condition or disease; relevant communities) to set research priorities carefully.
2. Conduct prefunding scientific design assessments and regulatory consultations to minimize waste and address uncertainties before making funding commitments Engaging funders and regulators early in a structured design review process can foster shared understanding, strengthen the justification for the trial, and further support more informed funding decisions	- Funder/sponsor - Investigator - Regulatory authority	Important because... It allows deliberation of the trial design before the approval process—which can be time-consuming and costly—and will reduce the likelihood of the trial failing or being uninformative. Having this step before funds are committed allows uncertainties to be addressed. Without this... Flaws in the trial design may go unchecked, leading to a trial i.e. funded, but ultimately fails to deliver. How this can be best done... Hold a presubmission meeting that brings together sponsors, funders, regulators, and when feasible—ethics reviews, with the aim of carrying out a structured review of the trial design and methodology. Any uncertainties can be addressed before funds are committed.
3. Engage patients and communities from the outset to build trust, ensure relevance, and strengthen recruitment and retention Engagement that begins early and is grounded in local context helps ensure trial questions reflect priorities that are important to patients and communities, interventions are acceptable, and participation is sustained through trust and shared ownership.	- Funder/sponsor - Investigator - Patient/community representatives	Important because... Through these interactions, trial hypotheses can be carefully crafted to address real priorities; interventions can be evaluated to ensure they are acceptable to those with lived experience. Trust-building is also important to help recruitment and retention of participants. Without this... Trials may fail to address priorities that are relevant to patients and communities or may implement interventions that are not acceptable or feasible from their perspective. This can lead to challenges in recruitment, retention, and overall trial success. How this can be best done... Engage patients and communities for collaborative partnerships that are grounded in local context from the earliest stage possible. The principles of research coproduction are helpful when thinking about building trusted partnerships: avoiding tokenism; including listening to all perspectives; respecting and valuing all input; reciprocity; and a balanced power dynamic. This needs time. Perspectives from patients and community can inform recruitment and retention strategies, including effective communication with participants. Embedding patient and community members in advisory boards and patient and public input (PPI) within trial protocols creates an enduring source of valuable input.
4. Apply structured design tools, established frameworks, and Open Science practices (eg, COMET, SPIRIT, pre-registration) to improve trial quality, transparency, and reproducibility Using structured tools and Open Science practices embeds consistency and accountability across the trial lifecycle and supports further	- Funder/sponsor - Investigator	Important because... The overall quality and transparency of the trial are upheld. Without this... There may be gaps, inconsistencies and uncertainties, deeming the findings less useful and contributing to research waste. How this can be best done... Use tools, frameworks, and guides from the earliest stage and throughout the trial lifecycle. Examples: Core Outcome Measures in Effectiveness Trials (COMET) Initiative; Enhancing the QUALity and Transparency Of health Research (EQUATOR) Network; Standard Protocol Items:

(Continued)

Table 1. Continued

Design	Key stakeholders	Rationale and implementation
coordinated design, conduct, and reporting		Recommendations for Interventional Trials (SPIRIT) checklist; PRO EDI; Consolidated Standards of Reporting Trials (CONSORT) statement; WHO Global Benchmarking Tool (GBT) for evaluation of national regulator systems for medicinal products. Follow OS best practice by preregistering studies or submit registered reports before data collection to ensure accountability and transparency.
5. Ensure accountability, particularly from funders or sponsors, for robust, justified trial designs, supported by multidisciplinary teams and collaborative networks that include statistical expertise • Coordinated efforts and diverse practical expertise enable more rigorous, well-justified trial designs, with funder and/or sponsor accountability ensuring that design choices and team capabilities are well supported	- Funder/sponsor - Investigator - Collaborative networks	Important because... It drives the development of rigorous, well-designed trials and ensures that teams have the capacity to deliver reliable, informative trials. Without this... Studies risk being underpowered, poorly designed, or misaligned with clinical needs, leading to wasted resources and exposing participants to unnecessary risks and burdens, with no clear expectation of scientific or social value from the resulting research. How this can be best done... Build collaborative networks and bringing together diverse expertise (eg, statistics, IT, clinical operations, and financial aspects). In circumstances where access to expertise is limited, it must not be left out (eg, in low resource setting; limited statistical expertise globally); funders can support buy-in of expertise.
6. Apply quality-by-design approaches targeting critical determinants of trial quality to strengthen design and reduce avoidable errors • A quality-by-design approach ensures that factors critical to trial quality are prospectively identified and managed to uphold trial integrity, participant safety, and the reliability of results	- Funder/sponsor - Investigator	Important because... Potential risks to quality are identified and mitigated during the clinical trial process. Without this... Avoidable errors may derail the clinical trial, put participants at risk, or result in findings that are unreliable and uninformative. How this can be best done... Plan a systematic and proactive strategy that uses risk assessment to identify potential sources of variability, and how these could impact the trial. Use monitoring activities to oversee aspects of the trial where there are potential risks. The European Clinical Research Infrastructure Network (ECRIN) hosts a toolbox of resources providing information on tools available for risk assessment, monitoring and study conduct, the institutions where they are used, and other relevant details such as links and user feedback.
Conduct	Key stakeholders	Rationale and implementation
7. Use future-proofing measures to safeguard trial integrity and ensure technologies remain relevant, reliable, and unbiased • Future-proofing involves anticipating changes in technology and research practice to maintain data integrity, minimize bias, and ensure that trial processes remain sustainable over time	- Funder/sponsor - Investigator	Important because... Integrity of the trial is safeguarded. Without this... There is a risk to data integrity and an increased risk of bias within the trial, which could compromise the validity of findings and expose participants to unnecessary risks, leading to wasted resources and diminished social value from the research. How this can be best done... Anticipate how technologies and practices might change during the trial. Use a “risk of bias” tool, such as the Cochrane Risk of Bias tool before the trial to identify and address areas of risk. The INSPECT-RS tool can be used for assessing trustworthiness of randomized controlled trials.
8. Assess feasibility through pilot studies, integration of qualitative evidence on lived experiences into trial process design, and engagement with local health professionals • A comprehensive approach to feasibility combines practical testing with contextual understanding, drawing on lived experience and	- Funder/sponsor - Investigator - Local health professionals	Important because... There is a clearer understanding of how delivering an intervention in a real-world setting will go. Without this... A full-scale trial could be funded and run, but lacking the learnings from a pilot study, there are many ways for the trial to fail. How this can be best done... Carry out pilot studies assessing practical aspects (eg, recruitment, adherence, acceptability, and logistics). Barriers and enablers can be identified, and the trial design refined in response to these. Take

(Continued)

Table 1. Continued

Conduct	Key stakeholders	Rationale and implementation
local expertise to anticipate barriers and refine trial delivery		account of lived experiences of people living with a disease or condition, including health care contexts and perceptions of usual care. Involve local health professionals to ground the pilot study in local context and to support effective recruitment and retention.
9. Provide continuous training, structured learning, and mentorship to build trial team competencies that go beyond current Good Clinical Practice (GCP) training • Building trial team capability requires ongoing, skills-based training and mentorship that translates GCP into consistent, high-quality conduct across all stages of a trial	- Funder/sponsor - Investigator - Professional bodies	Important because... Team competencies and capabilities are raised for delivery of a high-quality trial. Without this... There may be gaps in knowledge and missed opportunities, or risks which could violate the trial's integrity, undermining quality. How this can be best done... Plan for ongoing and structured learning for the team from the beginning and throughout the trial lifecycle. Plan for team members to develop skills with an emphasis on informed consent, compliance with protocols, and participant safety. Building skills around communication, cultural humility, inclusive practices and collaboration contribute to a broader strengthening of quality in trial delivery. Mentorship schemes can support professional development and the ability to apply knowledge and skills appropriately in the trial context.
10. Maintain oversight throughout the lifecycle of a trial through ethics review boards, trial steering committees, and data monitoring to uphold scientific rigor and support evidence-informed analyses • Effective oversight ensures that ethical standards are upheld, emerging risk are managed, and analyses are guided by credible evidence, maintaining confidence in both the trial process and its outcomes	- Funder/sponsor - Investigator - Ethics bodies - Trial oversight committees	Important because... Transparency and rigor are upheld in the research. Without this... There is a risk to the ethics and quality of the trial. How this can be best done... Ensure oversight from ethics review boards, trial steering committees, and a committee for data protection and monitoring. Where multisite trials cross jurisdictions, teams should aim to harmonize ethics submissions where possible to avoid duplication of effort. A trial steering committee can oversee that the trial is conducted in accordance with legislation, regulations, guidelines, and best practices and can advise on, or take action if there are potential research violations. A data protection committee can ensure compliance with relevant data protection laws and best practices to ensure robust data handling. Multisite trials may have different data considerations across jurisdictions, or unique contextual factors such as the use of indigenous data and knowledge.
Knowledge mobilization	Key stakeholders	Rationale and implementation
11. Ensure trial registration is monitored, and results are published in a timely and complete manner to maintain accurate records, meet ethical obligations, and reduce selective reporting and publication bias • Active monitoring of trial registration and reporting promotes transparency, protects research integrity, and ensures that all findings, regardless of outcome, contribute to the broader evidence base.	- Funder/sponsor - Investigator - Regulatory authority	Important because... Transparency and integrity are upheld. Without this... Trials may add to research waste. Failure to share and learn from research that does not produce a result in favor of the treatment under test has been recognized as a key source of clinical trial research waste and inefficiency. Selective or lack of reporting prevents the opportunity to build on previous research and can lead to duplication of effort. How this can be best done... Mandate that trials are registered prospectively, have a system for monitoring upkeep of records and reporting and ensure findings are published within a relevant timeframe.
12. Disseminate findings in accessible, inclusive formats, and mobilize knowledge to support equity and uptake into policy and practice	- Funder/sponsor - Investigator - Policymakers - Patient/community representatives	Important because... The results of the trial must reach all relevant audiences, and the information must be readily useable to inform policy. Without this... Changes may not be implemented, thereby rendering the trial wasteful, and it would be unethical to withhold the findings from relevant stakeholders.

(Continued)

Table 1. Continued

Knowledge mobilization	Key stakeholders	Rationale and implementation
Timely and inclusive dissemination ensures that trial evidence reaches all relevant audiences, supports equitable translation into policy, and enhances the real-world impact of the research.		How this can be best done... Carefully consider who needs to be served by the research, and who needs to be included in the trial. Research teams can refer to the PRO EDI tool to ensure a core set of characteristics, and additional characteristics as relevant. Ensure that everyone is proactively given the opportunity to know about the trial findings. Engage with policymakers, practitioners, and communities from early on helps to support uptake of the results into practice at local, national, and international levels.

interaction and alignment between industry representatives and policymakers/advisors but also within stakeholder groups, development of more pragmatic and flexible procedures, and application of clear and transparent eligibility criteria for facilitating actions” [9]. This also resonates with our good practice findings, where the aim of adaptability can be reinforced through early consideration of trial feasibility, coordinated oversight efforts to maintain trial integrity, and ongoing training that enables teams to respond effectively to evolving research environments.

Knowledge mobilization can largely be understood as “maximizing the scientific knowledge gained from the trial” or, in other words, knowledge mobilization turns research findings into practical actions [10]. Throughout our three research strands, the potential for knowledge mobilization was evident in both pretrial and post trial reporting as well as wider efforts to share and apply trial findings. Inconsistencies in proactively registering trials, transparently maintaining registry and reporting data, and a lack of post trial reporting are well-documented phenomena within health research that contribute to uninformative trials [11,12]. Failure to report the results of trials that were incomplete or failed to meet their primary outcomes further contributes to research waste [13,14]. A comprehensive understanding of the trial process, from design through completion, is essential for mobilizing research into equitable, actionable outcomes that inform policy and serve the populations for whom the research is intended.

Complying with our good practice actions may have resource implications. For example, working with communities to develop effective and inclusive recruitment and retention strategies (Action #3) requires appropriate funding. However, not all actions involve additional costs. Using structured tools such as SPIRIT at the design stage (Action #4) is unlikely to require additional resources; trial teams simply need to use them. Where additional funding is needed, funders should be willing to provide it to ensure a trial is informative. It is also important to recognize that lack of funding alone does not explain the prevalence of uninformative trials. Lack of awareness, insufficient methodological expertise within the trial teams, and the absence of structures to support or enforce

actions such as those we propose are likely to be at least as important.

5. Conclusions

This initial mapping approach provided a fuller understanding of trial informativeness by drawing on differing yet connected research outputs, offering a basis for future discussion and decision-making. The full mapping exercise, including areas of divergence and broader insights, is planned for publication in 2026. However, the overarching good practice actions identified to strengthen research systems merits standalone consideration as they address broader drivers of trial quality, ensuring that improvements are not limited to isolated studies but embedded across the trial ecosystem. Future work should therefore implement, evaluate, and refine these findings to ensure their relevance and practicality across varying trial environments and contexts.

CRedit authorship contribution statement

Sarah R. Prowse: Writing – original draft, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Miriam Brazzelli:** Writing – review & editing, Methodology, Conceptualization. **Jude Bain:** Writing – review & editing, Conceptualization. **Shaun Treweek:** Writing – review & editing, Supervision, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

There are no competing interests for any author.

Supplementary data

Supplementary data related to this article can be found at <https://doi.org/10.1016/j.jclinepi.2026.112274>.

Data availability

Data supporting this study are available in the cited publications and in the supplementary material accompanying this article.

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